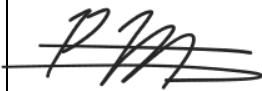


For use by the Project lecturer	Approved	Revision required
Feedback For use by the Project module lecturer only		

To be completed by the student					Language editor details		Language editor signature	
PROJECT PROPOSAL 2026				Project no	CF5	Revision no		0
Title	Surname	Initials	Student no	Study leader (title, initials, surname)				
Mr.	Male	PIM	23539102	Mr. Christo Fourie				
Project title (this is the title on the project concept note)								
Frequency Hopping Bluetooth Jammer								
Student declaration I understand what plagiarism is and that I have to complete my project on my own.						Study leader declaration This is a clear and unambiguous description of what is required in this project. Approved for submission (Yes/No)		
Student signature						Study leader signature and date		
						 Approved 2026-04-20		

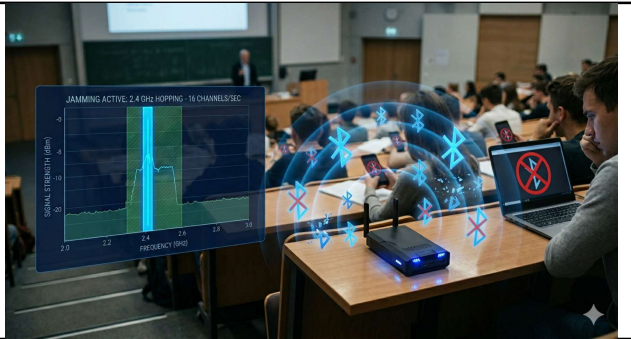
1. Project description.

What is the problem to be solved with your project? Describe in non-technical language what your project is about. What does your system have to do? Refer explictly to Figure 1 that you inserted on the right.

The problem addressed by this project is the use of Bluetooth audio voice communication devices to facilitate cheating during academic assessments or interviews. With the increase in availability of small and concealable Bluetooth communication devices (such as Bluetooth earphones) for transmitting answers via voice, existing methods of assessment invigilation are no longer sufficient.

There is thus a need for a small portable Bluetooth jamming device, which can cover a small group of students or an interview space and make voice transmitted or received by Bluetooth devices unintelligible, rendering them ineffective for communicating answers, as shown in Figure 1. To comply with legal requirements that limit transmission energy, this project aims to implement a portable narrow-band Bluetooth jamming device (that covers only a portion of Bluetooth’s hopping range) that shall make use of multiple frequency hopping jamming algorithms, implemented on an embedded platform, that can be swapped between to remain within legal transmission limits and maximise energy efficiency.

Figure 1. Insert an image below that captures the idea with your project. Explain in the text on the left.



2. Technical challenges in this project

Describe the technical challenges that are *beyond* those encountered up to the end of third year and in other final year modules.

2.1 Primary *design* challenges

What is the intended core technical challenge in the project? Which aspect or aspects of the design of the system do you expect to be the most challenging?

As the jammer is not a party to the Bluetooth connection, it has no access to the pseudo-random hopping sequence, and can cover only a fraction of the 80 MHz hopping range while having a limited legal isotropic radiated power of only 20 dBm (100 mW) [1]. As a result, the system must be designed to move its limited bandwidth around intelligently. The multiple jamming algorithms that shall be implemented to solve this problem pose significant design challenges in detecting the hop sequence and estimating the ideal jamming location while taking into account the system's delay and systems built into the Bluetooth hop sequence generator, which cause it to avoid known congested channels.

2.2 Primary *implementation* challenges

Which aspect or aspects of the implementation do you expect to be the most challenging?

The system has a significant timing challenge, as Bluetooth communication hops 1600 times per second. The communication between the software-defined radio (SDR) and the embedded platform, as well as algorithm speed, must be optimised to the extent that the different algorithms' output can be exploited within the small 625-microsecond window before communication hops again. The system must function fast enough that reception (detection) and transmission (jamming) occur within an exploitable window.

All processing will be done on an embedded platform, and not on a PC platform. Confirmed by having ticked this box.



Library functions will be used for hardware interfacing only. Confirmed by having ticked this box.



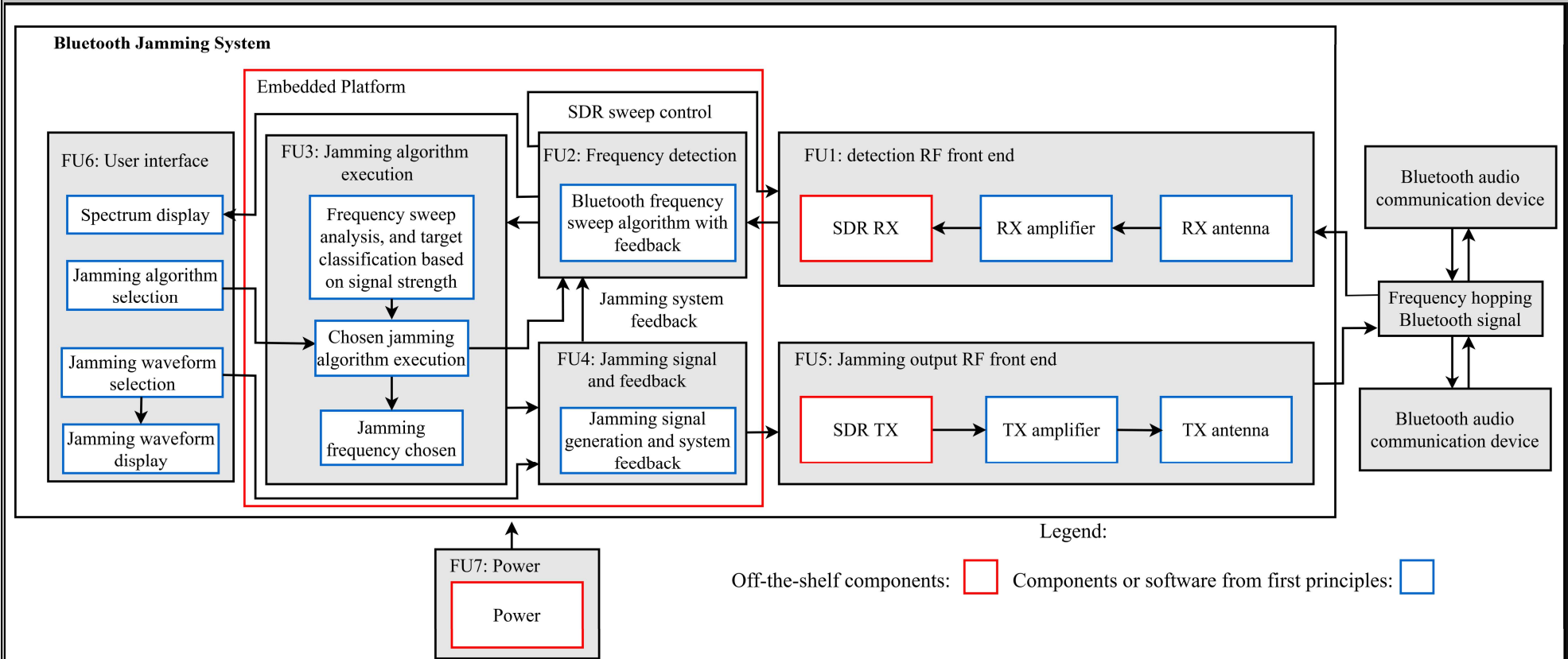
3. Functional analysis

3.1 Functional description

Describe the design in terms of system functions as shown on the functional block diagram in section 3.2. This description should be in *narrative format*. **DO NOT** use a bullet list.

The radio frequency (RF) detection front-end (FU1) shall consist of a directed detection antenna, which shall detect Bluetooth signals. This signal shall then be amplified before being fed into an off-the-shelf SDR receive (RX) component, which shall convert the signal into digital data for analysis by the embedded platform. A signal processing algorithm (FU2) shall use the RX information from the SDR to determine on which of the 79 Bluetooth bins information is being transmitted. It shall use feedback from itself and other parts of the system (FU3 and FU6) to control how it sweeps over the range to improve how quickly it can detect the signal. It will control where the SDR of FU1 sweeps to search for the Bluetooth hop. The spectrum information and the bin history shall be used by one of multiple intelligent algorithms (FU3), which shall be chosen by the user interface (UI) (FU6). This chosen algorithm shall determine the frequency to which the jamming signal must be tuned to maximise disruption to the Bluetooth communication. If necessary, the algorithm shall give feedback to FU2 to modify how the system is sweeping. The jamming signal and feedback control (FU4) shall take the algorithm output and chosen waveform by the UI and give the correct control to the SDR for it to generate a signal at the optimal frequency and chosen waveform at the correct time. If necessary, it shall give feedback to FU2 to modify how the system is sweeping. This SDR control shall be used by the RF transmission (jamming) front-end (FU5), which shall consist of an off-the-shelf SDR transmit (TX) component that shall generate the required signal. This signal shall be amplified by an output amplifier, which has a high noise tolerance (as jamming signals are noisy by design) and transmitted via a directed antenna. The UI (FU6) shall have 2 possible displays: the spectrum generated by FU2, and the chosen jamming waveform. This waveform, and the jamming algorithm can be chosen by the user (as multiple jamming algorithms shall be implemented). The power module (FU7) shall be an off-the-shelf module that provides power to the entire system.

3.2 Functional block diagram (this should not be a flow diagram)



4. System requirements and specifications

These are the core requirements of the system or product (the mission-critical requirements) in table format **IN ORDER OF IMPORTANCE**. Requirement 1 is the most fundamental requirement.

	Requirement 1: the <u>fundamental</u> functional and performance requirement of your project	Requirement 2 (Second in order of importance)	Requirement 3 (Third in order of importance)
1. Core mission requirements of the system or product. Focus on requirements that are core to solving the engineering problem. These will reflect the solution to the problem.	The jammer must have sufficient range and angular directivity to cover a medium-sized lecture hall when placed in the corner of the lecture hall.	Within the specified range, the system must disrupt enough Bluetooth transmission packets in order for speech transmission over Bluetooth to be unintelligible.	The jammer must not go over the limit of electromagnetic radiation in the ISM band as to remain legally compliant.
2. What is the <u>target specification</u> (this must be in <i>measurable</i> terms) to be met in order to achieve the requirement in 1. above?	The system shall have an effective jamming radius of 1.25 metres, this will result in a circle of diameter 2.5 metres around the system	The system shall disrupt 36% or more of the transmitted Bluetooth packets (packet delivery ratio of 0.64 or lower for the Bluetooth system) in the effective range, while the jamming system is active and using its most effective algorithm.	The system shall have an output jamming power of no more than 20 dBm.
3. Explain where this <u>specification originated</u> : i.e., explain the <u>value that you selected</u> . I.e., <i>how</i> will meeting the specification given in point 2 above <i>solve the problem</i> ?	This is the range by which 10 students seated in a standard medium sized university lecture hall at the University of Pretoria can be covered when seated close to each other. Engineering 1-3 was measured specifically.	Human speech becomes unintelligible if only 64% or less of the information of the communication is understood [2]. Thus disrupting 36% or more of packets shall make human speech over Bluetooth unintelligible.	20 dBm is the legal limit for electromagnetic radiation in the ISM band (2.400 GHz to 2.4825 GHz) [1] and cannot legally be exceeded without a licence.
4. How will you <u>demonstrate at the examination</u> that this requirement and specification (points 1 and 2 above) have been met? Be explicit about how you will prove these were met.	This shall be demonstrated by showing that a Bluetooth earphone has its packet delivery ratio reduced to at most the specified value (0.64) when placed within the cone of effective range.	An electromagnetic (EM) receiver placed in the jammer's effective range shall measure that a jamming signal overwhelms at least 36% of the Bluetooth transmission packets observed by the receiver over 15 seconds when the jammer is on.	The EM transmission power of the system shall be measured by the Compact Range at the University of Pretoria. This will provide a detailed analysis of the transmission power of the system.
5. Your own design <u>contribution</u>: what are the aspects that <i>you will design and implement yourself</i> to meet the requirement in point 2? If none, <u>remove this requirement</u> .	The antenna and amplifier must be designed to have have the gain and directivity specifications necessary to satisfy the range and angular requirements.	The antenna and amplifier must generate significant enough jamming power to jam Bluetooth hops, while the algorithms must perform well enough to achieve the packet delivery ratio.	The antenna and amplifier must be designed to have have the gain and directivity specifications necessary to ensure radiation is legally compliant.
6. What are the aspects to be <u>taken off the shelf</u> to meet this requirement? If none, indicate "none". Clearly specify for what tasks library functions will be used (if relevant to the project).	An off-the-shelf software defined radio shall be used to generate the initial jamming signal based on commands from an embedded platform.	All processing shall be performed on an off-the-shelf embedded platform. An off-the-shelf software defined radio shall be used to generate the initial jamming signal. Basic mathematics, hardware interfacing and FFT software libraries will be used.	none

System requirements and specifications page 2

	Requirement 4 (Fourth in order of importance)	Requirement 5	Requirement 6
1. Core mission requirements of the system or product. Focus on requirements that are core to solving the engineering problem. These will reflect the solution to the problem.	The system must be able to saturate the entirety of a Bluetooth bin in order to jam that specific hop's content.	The system must be able to shift the jamming signal over the entirety of the range where Bluetooth hops to, such that each of the 79 bins can be jammed.	
2. What is the target specification (this must be in <i>measurable</i> terms) to be met in order to achieve the requirement in 1. above?	The output jamming signal must have a bandwidth of no less than 1 MHz, and no greater than 20 MHz.	The output jamming signal must be able to hop from 2.402 GHz to 2.480 GHz.	
3. Explain where this specification originated: i.e., explain <u>the value that you selected</u> . I.e., <i>how</i> will meeting the specification given in point 2 above solve the problem?	Bluetooth communication bins have a bandwidth on 1 MHz and the jammer must be able to cover the entirety of a single bin. While a frequency band which is too wide would trivialise the project.	The 79 Bluetooth communication bins are spread from 2.402 GHz to 2.480 GHz [3]. The jammer must be able to hop across this entire range to jam every bin Bluetooth will use.	
4. How will you demonstrate at the examination that this requirement and specification (points 1 and 2 above) have been met? Be explicit about how you will <i>prove</i> these were met.	This shall be shown on a spectrum analyser. The analyser's probes shall be set 1 MHz, apart for the frequency of intermediate (2.441 GHz) Bluetooth hop bin [3], and the output signal must be at least as large as this range, and smaller than 20 MHz.	This shall be shown on a spectrum analyser. The system must produce a peak at the greatest (2480), smallest (2.402 GHz), and intermediate (2.441 GHz) Bluetooth hop bin [3], which will cover the entirety of the specified range.	
5. Your own design contribution: what are the aspects that <i>you will design and implement yourself</i> to meet the requirement in point 2? <u>If none, remove this requirement.</u>	The antenna and output amplifier, which shall be designed and implemented from first principles must have an instantaneous bandwidth of 1 to 20 MHz, and must thus be designed to meet this bandwidth requirement.	The antenna and output amplifier, which shall be designed and implemented from first principles must maintain performance across the entirety of the specified frequency range.	
6. What are the aspects to be taken off the shelf to meet this requirement? If none, indicate "none". Clearly specify for what tasks library functions will be used (if relevant to the project).	An off-the-shelf software defined radio shall be used to generate the initial jamming signal based on commands from an embedded platform.	An off-the-shelf software defined radio shall be used to generate the initial jamming signal based on commands from an embedded platform.	

System requirements and specifications page 3

	Requirement 7 (Seventh in the order of importance)	Requirement 8	Requirement 9
1. Core mission requirements of the system or product. Focus on requirements that are core to solving the engineering problem. These will reflect the solution to the problem.			
2. What is the target specification (this must be in <i>measurable</i> terms) to be met in order to achieve the requirement in 1. above?			
3. Explain where this specification originated: i.e., explain <u>the value that you selected</u> . I.e., <i>how</i> will meeting the specification given in point 2 above solve the problem?			
4. How will you demonstrate at the examination that this requirement and specification (points 1 and 2 above) have been met? Be explicit about how you will <i>prove</i> these were met.			
5. Your own design contribution: what are the aspects that <i>you will design and implement yourself</i> to meet the requirement in point 2? If none, <u>remove this requirement</u> .			
6. What are the aspects to be taken off the shelf to meet this requirement? If none, indicate "none". Clearly specify for what tasks library functions will be used (if relevant to the project).			

5. Real world field conditions

These are the ACTUAL, REAL WORLD CONDITIONS under which your project has to work *and will be demonstrated*.

	Real world field condition 1 (required to include)	Real world field condition 2 (optional to include)	Real world field condition 3 (optional to include)
Field condition requirement. In which real world conditions does the system have to operate?E.g., "lab conditions" or "outside in uncontrolled conditions". Describe the one, two or three most important field conditions.	The system must work in a typical university lecture hall, with no shielding from electromagnetic radiation or noise that may be present.	No WiFi devices shall be operating in the lecture hall, as these shall not be present in exam conditions.	

6. Student tasks

6.1 Design and implementation tasks of the student

List your primary design and implementation tasks in bullet list format (5-10 bullets). These are *NOT* product requirements, but *YOUR* tasks.

- The student shall design two high-frequency (greater than 2.402 GHz) amplifiers, one for jamming signal transmission and one for Bluetooth signal detection.
- The student shall design two high-frequency (greater than 2.402 GHz) antennas, one for jamming signal transmission and one for Bluetooth signal reception. Both must conform to the directivity and angular requirements.
- The analogue components of the project shall be implemented on a printed circuit board, and verified to work correctly when connected to the off-the-shelf SDR.
- The student shall design multiple jamming frequency shifting algorithms that can be selected by the user. The algorithms will be compared based on jamming performance.
- The student shall implement a user interface that shall be used to select the jamming algorithm and jamming signal, and display the frequency sweep and jamming waveform of the system.
- The student shall increase the speed of the designed algorithm while implementing any other possible time speedups in code or communication to reduce the system's reaction time as much as possible.

6.2 New knowledge to be acquired by the student

Describe what the theoretical foundation to the project is, and which *new* knowledge you will acquire (*beyond* that covered in any other undergraduate modules).

- The student shall obtain knowledge on frequency hopping communication in order to understand the Bluetooth hopping sequence and other communication principles, such as processing gain.
- The student shall obtain knowledge on communication jamming principles, especially hopping jammers, such that the jammer's overall design is implemented in the optimal manner.
- The student shall obtain knowledge on high-frequency RF electronic design in order to design and build adequate high-frequency amplifiers for transmission and reception with the unique challenges of signal integrity and potential unwanted EM radiation.
- The student shall obtain knowledge on detection and estimation theory such that a predictive algorithm for the Bluetooth hopping sequence can be implemented, and the optimal choice of algorithms to be implemented for the specific task are chosen.

7. References

Format your reference list either in IEEE reference format or in APA reference format. Check what your study leader prefers and use consistently.

- [1] D. Cull and S. Makda, "Guide: Commonly-used Licence-exempt bands in South Africa which may be used for outdoor wireless access systems," Ellipsis Regulatory Solutions, Cape Town, South Africa, May 2017. [Online]. Available: <https://www.ellipsis.co.za/wp-content/uploads/2018/04/Guide-to-commonly-used-licence-exempt-frequency-bands-May-2017-1.pdf>
- [2] D. Crowther et al., "The relationship between intelligibility and comprehensibility in second language speech," *Bilingualism: Language and Cognition*, Cambridge University Press, Sep. 2025. [Online]. Available: <https://doi.org/10.1017/S1366728925000173>
- [3] Bluetooth Special Interest Group, Bluetooth Core Specification, Version 6.0, Bluetooth SIG, Kirkland, WA, USA, Feb. 2024. [Online]. Available: <https://www.bluetooth.com/specifications/specs/core60-html/>